PowerODBA Ansible Collection

Overview: The Power Oracle Database Automation (PowerODBA) Collection modules are based on the Oravirt collection https://github.com/oravirt/ansible-oracle which helps automation of Oracle Database Administration activities on AIX. These have been modified and tested exclusively to work on AIX.

Capabilities:

- Database creation [Single Instance/RAC & Multitenant]
- Apply RU Patches [Standalone DB/Database on ASM & RAC]
- Manage Users [Create/drop users, grant/revoke privileges]
- Manage Pluggable databases.
- Manage Tablespaces.
- Manage Redo logs.
- Manage Database directories.
- Manage ASM
- Manage ACFS
- Manage DBMS jobs

  PODBA v2.0 and later: The following automation capabilities have been introduced
- Upgrade Single Instance Grid & Databases from 12c to 19c.
- Upgrade RAC Grid & Databases from 12c to 19c.
- Implement Physical Standby.
- Download Patches from My Oracle Support.
- Oracle 19c cross-endian database migration from x86 Linux to AIX using Oracle's M5 method [RMAN incremental backups & Full Transportable Data Pump]

Version Change History:

- **PODBA v3.0.0:**
  - Migrated all the Oracle modules for Ansible from legacy cx_Oracle to modern renamed successor python-oracledb.
  - Enhanced database and grid facts gathering modules (oracle_facts with 9 comprehensive categories including OPatch inventory, tablespaces, redo, and jobs; oracle_gi_facts with OPatch patches, crsctl resource status, and asmcmd lsdg diskgroup metrics)
  - Modernized documentation for all the migrated modules with structured Markdown guides, verified execution logs, and SQL/CLI verification commands.
  - Introduced Oracle 19c cross-endian migration from x86 Linux to AIX using Oracle's M5 method and the DBMIG 5.1 scripts (MOS Doc 2999157.1). Supports Non-CDB to Non-CDB, PDB to PDB and Non-CDB to PDB.

- New playbook m5_migration.yml with roles mig_prechecks, mig_rman_L0_L1 and mig_rman_L1F, run as tagged stages: prechecks, L0 baseline, repeatable L1 incrementals, L1F cutover and go-live.
- Migration hosts are defined in their own [mig] inventory group (mig_source, mig_destination), so migration runs never touch other hosts.
- Added ansible.posix as a collection dependency.

- **PODBA v2.0.9:**
  - Tested role separation during upgrade.
  - Fixed the bug in oracle_opatch module, ansible interpreter is using the shemang python. Removed shebang from oracle_opatch module.
  - Upgrade module will import oracle install and gi install roles from Oracle Single Instance collection.
- **PODBA v2.0.8:** Refactored oradb_create role for integration with Oracle SI ansible collection
- **PODBA v2.0.7:** Introduced Dataguard feature for single instance database using RMAN Backup and RMAN DUPLICATE method.
- **PODBA v2.0.6:** Introduced RAC Grid Infrastructure and Database Upgrade from 12c to 19c.

  Following roles are added.
  a) podba_rac_upgrade_precheck
  b) podba_rac_crs_upgrade
  c) podba_rac_db_install
  d) podba_rac_db_upgrade

- **PODBA v2.0.5:**
  - Latest RU patches can be applied during the upgrade to 19c. Previously it was restricted till RU 19.17.
  - Introduced patch staging options – NFS, local (target node) and remote (ansible controller) for Single Instance Upgrade.

- **PODBA version 2.0.4:** o Enhanced the role “oraswdb_manage_patches” patching module which uses “opatch” utility with restart of database and listener services while patching.
  - Added support for patch staging options – NFS, local (target node) & remote (ansible controller).

- **PODBA version 2.0.3:** Refactored variables to a common path for easy usage.

- **PODBA version 2.0.2:** Enhanced the roles oracle_install & db_upgrade to upgrade multiple Oracle databases to 19c.

- **PODBA version 2.0.1:** Added new role orasw_download_patches to Download Patches from My Oracle Support.

- **PODBA version 2.0.0:** Added three roles to upgrade Single Instance Grid & Database from 12c to 19c.
    a) podba_si_has_upgrade
    b) podba_oracle_install
    c) podba_db_upgrade

- **PODBA version 1.0.1**: Provided a single file to store all the user passwords making it feasible to use "ansible-vault" for security.

- **PODBA version 1.0**: Initial release.

Assumptions:

1. The user is familiar with Ansible and has basic knowledge on YAML, for the purpose of running this playbook.
2. The user is familiar with Oracle Database Administration.
3. The user is familiar with the AIX Operating system.
4. The version of AIX is 7.2 TL4 SP1 or later. (It should work on other versions of AIX supported by the oracle database AIX OS requirements but has not been tested).
5. The DB version which was tested is 19c and it may or may not work on other DB versions.

Getting started:

To get started with Ansible refer:

https://docs.ansible.com/ansible/latest/user_guide/intro_getting_started.html To get started with Oracle Database on AIX refer:

https://docs.oracle.com/en/database/oracle/oracle-database/19/axdbi/index.html

https://www.ibm.com/support/pages/oracle-db-rac-19c-ibm-aix-tips-and-considerations

To get started with AIX refer:

https://www.ibm.com/support/knowledgecenter/ssw_aix_72/navigation/welcome.html

Prepare the Ansible Controller:

The following will describe the setup of Ansible environment for command line. To use Ansible Automation Platform (GUI), please refer https://github.com/IBM/ansible-poweraixoracledba/blob/main/docs/PowerODBA_using_AAP2.pdf

1. Requires Ansible >= 2.10 on Linux on Power (or) x86 machine.
2. Requires Python 3.6 (or) later [dnf install python3].
3. **oracledb**: This is a python module which makes the connection to Oracle database using a database user. Example: sys, system or any non-system user.

   Prerequisites: gcc, python3x-devel [dnf install gcc python3x-devel] (x is the version of python. Ex: If python version is 3.9.16, use 'dnf install python39-devel')

   Online installation:
   As root: python -m pip install oracledb --upgrade
   As a non-root user: python -m pip install oracledb --upgrade --user

4. **Install the collection:** Run the following command.
   a. $ ansible-galaxy collection install ibm.power_aix_oracle_dba

Documentation: The detailed steps for each capability can be found here:

1. PowerODBA_Ansible_Collection.pdf : This file.
2. PowerODBA_using_AAP2.pdf : Playbooks execution using GUI.
3. ASM Create Diskgroups
4. ASM Add Disks
5. ASM Drop Disk
6. ASM Drop Diskgroup 7. Manage SQL Scripts
8. Manage SQL Queries
9. Create Databases
10. View DB Facts
11. Manage Database Directories
12. Manage DBMS Jobs
13. Delete Database
14. View GI Facts

15. Gather Global Statistics
16. Manage AWR
17. Manage Initialization Parameters
18. Manage ACFS on RAC
19. Manage ACFS
20. Manage Database Grants
21. Manage Job Class
22. Manage Job Schedule
23. Manage Job Window
24. Manage PDBs
25. Manage REDO
26. Manage Database Roles
27. Manage Tablespaces
28. Manage Users
29. Patch Standalone GI Home
30. Patch RAC GI Home
31. Patch Standalone RDBMS Home
32. Patch RAC RDBMS Home
33. Patch Download
34. Run Datapatch
35. Manage Resource Consume Groups
36. Upgrade Single Instance Grid and Databases: Oracle Single Instance Grid & Databases upgrade.
37. Upgrade RAC Grid and Databases: Oracle RAC Grid Infrastructure and Databases upgrade.
38. Single Instance Dataguard Setup: Physical Standby Database configuration for Single Instance database.
39. Oracle Cross Platform Migration: Oracle 19c cross-endian database migration from x86 Linux to AIX using Oracle's M5 method [RMAN incremental backups & Full Transportable Data Pump]